

Computational Geometry

MCS 481, CRNs: 40620, 4621

3 undergraduate hours, 4 graduate hours

Fall 2026, MWF 12pm-12:50am, Lincoln Hall 215

About Your Instructor

Instructor Name: Jan Verschelde (You may call me Professor Jan.)

Office: SEO 1210

Office Phone: 312 996 4609

Drop-in (Office) Hours: From 2pm to 3pm on Mondays, Wednesdays, and Fridays; or by appointment.

Virtual via <https://uic.zoom.us/my/profjanofficehour>.

Email: janv@uic.edu.

Methods of Instruction: The lectures happen in Lincoln Hall 215.

Course Description

Computational Geometry is devoted to the study of algorithms to solve geometrical problems.

The development of the field is driven by applications to computer graphics, computer aided design, motion planning and visibility problems in robotics, geographic information systems, integrated circuit design, etc...

What You Will Learn in This Course

Course Goals and Student Learning Outcomes

The goals of MCS 481 are to understand the statements and properties of geometric problems; to know algorithmic techniques for solving geometric problems; and to apply software for exploring geometric algorithms.

Pre- or Co-requisites For This Class

Grade of C or better in MCS 401; or consent of the instructor.

Course Website

The course web site is at <https://homepages.math.uic.edu/~jan/mcs481> and a backup site is at <https://janverschelde.github.io/mcs481/>

On the course web site, the syllabus, lecture slides and homework assignments are posted. In this course, there will be some minimalistic use of canvas. Gradescope will be used to collect all answers to homework and exams.

Course Texts and Materials

The textbook for this course is *Computational Geometry. Algorithms and Applications* by M. de Berg, M. van Kreveld, M. Overmars, and O. Schwartzkopf, Springer-Verlag. Slides used in the lecture are based on the textbook, which provides an extended narrative.

Required Technology

As the project requires the use of software, you should have access to a computer.

Required Reading, Video, or Lab Materials

Pre-recordings of the Fall 2023 lectures were posted at youtube, available as a playlist at <https://www.youtube.com/playlist?list=PLGMCOM5grOW6W532C12AemiZZfY6X1x2Q>

These videos are still relevant, although the current lectures may have some substantial modifications.

Our Inclusive Learning Environment

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If there are aspects of the instruction or design of this course that result in barriers to your inclusion, engagement, accurate assessment or achievement, please notify me as soon as possible.

Disability Accommodation Procedures

*UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, you should connect with the **Disability Resource Center (DRC) at drc.uic.edu, drc@uic.edu, or at (312) 413-2183** to create a plan for reasonable accommodations. In order to receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Letters of Accommodations can be presented at any point of the semester and, unless otherwise noted, do not expire once obtained. Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.*

Community Agreement

Community Agreement: Ground Rules for a Safe/Brave Space

- *Be present (turn off cell phones and remove yourself from other distractions)*
- *Be respectful*
- *No food and no snacking*
- *No side conversations*
- *Be willing to work together*
- *Be mindful of one another's privacy -- do not invite outsiders into our classroom*

Students who engage in any behavior that results in the disruption of a class may be directed to leave the classroom for the remainder of the class period.

How Your Learning Will Be Assessed

There will be several homework collections, two midterm exams, three projects, and a final exam.

Selected homework problems are assigned with each lecture but is recommended that you try all exercises. Homework will be collected every week, except during exam weeks. The main goal of the lectures is to get you started with the solution of those problems. All homework counts for 100 points.

Three projects will be assigned during the semester, worth a total of 200 points.

During the semester, there will be two midterm exams, each worth 100 points. No makeup for the midterm exam will be given. If the midterm exam is missed, then greater weight will be placed on the final exam, especially on the material covered on the missing exam.

The final exam counts for 200 out and is mandatory for passing the course. Students may replace the final exam with a research computer project combined with a written technical report.

Assessments

Your course grade is based on a total of 700 points: 100 from the homework, 200 from the projects, 200 from the midterm exams during the semester, and 200 from the final exam.

Grading scale

A	90% - 100%
B	80% - 89%
C	70% - 79%
D	60% - 69%
F	0% - 59%

Student Accountability for Coursework

When requesting an extension of a deadline to submit homework, you may expect to receive a couple of hours, not days or a week. Submitting a partial answer to homework is better than submitting nothing.

Midterm Grades

There is no formal midterm grade. By the day of the midterm exam, after four weeks of classes, you should have a good understanding of first half of the materials and to a good grade in the course.

If after four weeks of classes you still feel lost in the course, then please reach out to me for help.

Final Exam

The final exam is cumulative, covering all topics in the course.

Course Policies

Tips for Success

When learning a new algorithm as a computer scientist ask yourself the following question: can I write detailed instructions for a computer to execute this new algorithm? While you will not be asked to produce working code in this course, asking this main question should guide your learning of any new algorithm.

There are many ways for you to show us what you know or can do and how you are learning, through your effort, interactions, and class participation, application of critical thinking skills to solve real world problems, and performance on assignments and exams. This class has been structured to help all students get the support and guidance needed to succeed in your learning. The following tips will help you be successful in this class and in other classes throughout your career at UIC.

- *Ask questions during class.*
- *Complete assignments on time.*
- *Participate in class discussions.*
- *Individually review your notes after class.*
- *Post questions on the discussion board or ask questions during drop-in (office) hours about concepts or procedures that seemed confusing.*

Finally, do the exercises! They are the low-stakes assignments in this course. Many of the exercises help you to become familiar with the algorithms and prepare you for the high-stakes exams.

Read the textbook. The main goal of the lectures is to provide you with better access to the literature.

Attendance

Students are expected to attend all class meetings. Any changes in this syllabus or in the scheduling of exams and other assignments will be announced during class meetings.

Religious Holidays

I will make every effort to avoid scheduling exams or requiring student projects be submitted on [religious holidays](#). If you wish to observe your religious holidays, please notify me by the tenth day of the semester of the date when you will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, please notify me at least five days in advance of the date when you will be absent. I will make every reasonable effort to honor your request and not penalize you for missing the class. If an examination or project is due during your absence, you will be given an exam or assignment equivalent to the one completed by those students in attendance. Students may appeal through [campus grievance procedures](#) for religious accommodations.

Class Participation / Engagement

Because active participation in the course is one of the most important ways to learn, you will be assessed on your class engagement. Engagement looks a little different for everyone, but in general, an engaged student will come to class prepared, contribute regularly to class activities or discussions, listen attentively to peers and the instructor, stay on-task during class, complete their work in a timely manner, and reach out to the instructor if they have questions or start to fall behind. If you anticipate any barriers

to your full engagement in the course, I encourage you to contact me so we can strategize about how you can best fulfill the course requirements.

Your engagement or lack thereof will not be assessed.

Policy for Late or Missing Work

Deadlines may be postponed. If you know you will be late, then it is better to apply for an extension of the deadline, instead of not submitting anything. For missed assignments, greater weight may be placed on the final exam.

Policy for Revisions

You will not be allowed to resubmit revised homework for a better grade.

Policy for Regrading

Any request for a regrade must be made within one week of the assignment being returned to you. If you think there has been a simple arithmetic error on your assignment then write a note explaining the error, attach this to the front of the assignment, and turn it in. If the error is confirmed, then the points will be added to your score before the end of the semester. If you think there was an error in grading that is not an arithmetic error, write a brief note explaining why you think more points should be awarded, include it with the assignment, and submit it. If the points you request will affect your final grade, then I will reevaluate the assignment for the contested points.

Academic Integrity

No student shall claim or submit the work of another as one's own.

Allowing someone to copy from you is also punishable. All your work in this class will be individual. Verbatim copying the output of generative AI tools is plagiarism.

UIC is an academic community committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. In this community, all members including faculty, administrators, staff, and students alike share the responsibility to uphold the highest standards of academic honesty and quality of academic work so that such a collegial and productive environment exists.

As a student and member of the UIC community, you are expected to adhere to the [Community Standards](#) of integrity, accountability, and respect in all of your academic endeavors. When [accusations of academic dishonesty occur](#), the Office of the Dean of Students investigates and adjudicates suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct. Please review the [UIC Student Disciplinary Policy](#) for additional information about the process by which instances of academic misconduct are handled towards the goal of developing responsible student behavior.

By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not

communicated with anyone in any way during an exam or other online assessment. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

Use of Generative Artificial Intelligence

In response to my query "use of generative ai to computer algorithm" the AI overview of Google started its reply with "Using generative AI is a highly effective way to learn computer algorithms because it acts as an interactive, 24/7 personalized tutor." giving among its references an insightful 2024 paper in the Communications of the ACM on "Computing Education in the Era of Generative AI."

Using generative AI to help you learn is good as long as you learn, that is: you gain understanding and insight in the design and analysis of algorithms. Be aware of the disclaimer: "AI can make mistakes, so double-check responses."

FERPA

Federal law (FERPA) prohibits me from disclosing or discussing your grades over e-mail. Please come see me during drop-in hours or make an appointment if you want to talk about your work.

Course Schedule

Below is the outline of the course. The section numbers refer to the textbook.

Week 1 – chapter 1: computational geometry, an introduction

- L-1 Mon 24 Aug: definition and examples -- convex hulls, a first algorithm -- 1,2,3 (1.1)
- L-2 Wed 26 Aug: second convex hull algorithm -- degeneracies and robustness -- 4,7,9 (1.1-4)
- L-3 Fri 28 Aug: software for computational geometry

Week 2 – Chapter 2: Line Segment Intersection, thematic map overlay

- L-4 Mon 31 Aug: line segment intersection -- plane sweep algorithms -- 1,2,3 (2.1)
- L-5 Wed 2 Sep: output sensitivity -- event queues -- doubly connected edge lists -- 4,5,6 (2.2)
- L-6 Fri 4 Sep: overlays of subdivisions -- boolean operations -- 7,8,9 (2.3-5)

Week 3 – Chapter 3: Polygon Triangulation, guarding an art gallery

- Mon 7 Sep: Labor Day. No classes.
- L-7 Wed 9 Sep: triangulations and colorings -- the art gallery theorem -- 1,2,3 (3.1)
- L-8 Fri 11 Sep: partitioning a polygon into monotone pieces -- 5,6,8 (3.2)
- L-9 Mon 14 Sep: triangulating a monotone polygon -- 10,13,14 (3.3-4)

Week 4 – Chapter 4: Linear Programming, manufacturing with molds

- L-10 Wed 16 Sep: geometry of casting -- half-plane intersections -- 2,4,5 (4.1,2)
- L-11 Fri 18 Sep: incremental and randomized linear programming -- 10,11,12 (4.3,4)
- L-12 Mon 21 Sep: unbounded linear programs -- LP in higher dimensions -- 13,15,16 (4.5,6)

Week 5 – Chapter 5: Orthogonal Range Searching, querying a database

- L-13 Wed 23 Sep: range searching -- Kd trees -- 1,2,3 (5.1,2)
- L-14 Fri 25 Sep: range trees in dimensions two and higher -- 6,8,9 (5.3,4)
- L-15 Mon 28 Sep: general set of points -- fractional cascading -- 10,11,1 (5.5-7)

Week 6 – review and first midterm exam

- L-16 Wed 30 Sep: review of the first five chapters
- L-17 Fri 2 Oct: the first midterm exam covers chapters 1 through 5

Week 7 – Chapter 6: Point Location, knowing where you are

- L-18 Mon 5 Oct: point location and trapezoidal maps -- 1,3,4 (6.1)
- L-19 Wed 7 Oct: randomized incremental algorithm -- 5,7,8 (6.2)
- L-20 Fri 9 Oct: dealing with degenerate case -- a tail estimate -- 11,13,15 (6.3,4)

Week 8 – Chapter 7: Voronoi Diagrams, the post office problem

- L-21 Mon 12 Oct: Voronoi diagrams, definition and basic properties (7.1)
- L-22 Wed 14 Oct: computing the Voronoi diagram -- 4,6,8 (7.2)
- L-23 Fri 16 Oct: Voronoi diagrams of line segments -- farthest points -- 9,10,11 (7.3,4)

Week 9 – Chapter 8: Arrangements and Duality, supersampling in ray tracing

- L-24 Mon 19 Oct: supersampling -- computing the discrepancy -- duality -- 1,2,3 (8.1,2)
- L-25 Wed 21 Oct: duality -- arrangements of lines -- 5,6,9 (8.2,3)
- L-26 Fri 23 Oct: levels and discrepancy -- 10,12,13 (8.4,5)

Week 10 – Chapter 9: Delaunay Triangulations, height interpolation

- L-27 Mon 26 Oct: triangulations of point sets -- the Delaunay triangulation -- 1,2,3 (9.1,2)
- L-28 Wed 28 Oct: computing the Delaunay triangulation -- 5,6,8 (9.3)
- L-29 Fri 30 Oct: analysis -- a framework for randomized algorithms -- 9,11,16 (9.4,5)

Week 11 – Chapter 10: More Geometric Data Structures, windowing

- L-30 Mon 2 Nov: more geometric data structures: interval trees -- 1,4,9 (10.1)
- L-31 Wed 4 Nov: priority search trees -- 2,3,10 (10.2)
- L-32 Fri 6 Nov: segment trees -- 5,7,8 (10.3)

Week 12 – review of chapters 6 tot 10, second midterm exam

- L-33 Mon 9 Nov: the complexity of convex hulls in 3-space -- 5,6,7 (11.1)
- L-34 Wed 11 Nov: review of chapters 6 to 10
- L-35 Fri 13 Nov: the second midterm exam covers chapters 6 through 10

Week 13 – Chapter 11: Convex Hulls, mixing things

- L-36 Mon 16 Nov: computing convex hulls in 3-space -- 5,6,7 (11.2)
- L-37 Wed 18 Nov: analysis -- convex hulls and half-space intersection -- 8,9,10 (11.3,4)

Week 14 – Binary Space Partitions (Chapter 12) – Robot Motion Planning (Chapter 13)

- L-38 Fri 20 Nov: binary space partitions and the painter's algorithm -- 1,2,3 (12)
- L-39 Mon 23 Nov: robot motion planning -- Minkowski sums -- 1,4,5 (13)
- Wed 25 Nov: Student Wellness Day. No classes.
- Fri 27 Nov: Thanksgiving holiday. No classes.

Week 15 – review for final and/or project presentations

- L-40 Mon 30 Nov: review of chapters 1 through 5 and/or project presentations
- L-41 Wed 2 Dec: review of chapters 6 through 10 and/or project presentations
- L-42 Fri 4 Dec: comprehensive review and/or project presentations

Academic Deadlines

Important dates in the semester are listed below.

- Wednesday 23 September: Project one is due at noon.
- Friday 2 October: first midterm exam.
- Wednesday 21 October: Project two is due at noon.
- Friday 13 November: second midterm exam.
- Wednesday 2 December: Project three is due at noon.
- Week of 7 December: final exam, date, time, and room to be announced.

Providing Feedback About This Course

At the end of the term, you will be asked to complete the course evaluations.

UIC Resources - Supporting Student Health, Wellbeing, and Academic Success

As a student at UIC, you may experience challenges such as struggles with academics, finances, student life, or your personal well-being. Please know this is completely normal and that you shouldn't hesitate to ask for help. Come to me, or if it is about an issue beyond the scope of this class, please contact your college advisors, or get help from any number of other support services and resources available to all UIC students:

- [Current Student Resources](#) – a one-stop shop for links to resources in the following categories: General, Academic, Student Support, Student Life, Technology, Health and Safety, and Getting Around Campus
- [UIC Tutoring Resources](#)
- Programs and initiatives supporting the UIC undergraduate experience and academic programs through the [Student Success Initiative](#)
- If you are experiencing personal hardships (including but not limited to: personal and family emergencies, academics, interpersonal conflicts, personal safety, physical and mental health concerns, food or housing insecurity) and need resources and assistance, please reach out to the [Student Assistance](#) division, supported by UIC's Office of the Dean of Students.
- [Student Guide for Information Technology](#) – a comprehensive resource for UIC students describing the most commonly used IT services and tools supporting your success
- [Navigating Health Resources at UIC](#) – explore the many resources that are on campus to support students' health and wellbeing and learn about how you can access them.

Importantly, if you are in immediate distress, please know the UIC Counseling Center has crisis intervention services available 24 hours a day, 7 days a week. You can walk into the Counseling Center during business hours to request to talk to a crisis counselor, or you can call the Counseling Center main line at (312) 996-3490. If you are calling outside of business hours, select "2" during the automated greeting to be connected to a crisis counselor. You can find additional mental health resources and services on [the Counseling Center Website](#) and the [Student's Guide to Accessing Behavioral Health Services at UIC](#).

For immediate crisis support outside of UIC, call or text "9-8-8" any time, 24 hours a day, 7 days a week. Dialing 988 will connect you to a trained crisis professional. 988 offers services in Spanish, American Sign Language, and over 240 languages through tele-interpretation services. [The 988 website](#) has an online chat feature, too.

*The **Math and Science Learning Center**, located in the Science and Engineering South Building (SES) at 845 W. Taylor St. 3rd Floor, Room 247, is a meeting place for students in Math, Biological Sciences, Chemistry, Earth and Environmental Sciences, and Physics. At the MSLC, students can meet with graduate teaching assistants for tutoring in 100-level courses, attend informal group study sessions with other students, or meet up with friends to attend one of the workshops, seminars, or other activities*

sponsored by the MSLC during the semester. Visit the [MSLC website](#), call 312-355-4900, or email at mslc@uic.edu.

The **UIC Library** is located both on east and west campus, provides access to resources, study rooms, and research support both online via chat and in person. At Daley Library on the east side of campus, stop by the reference desk in the IDEA Commons, or make an appointment for research help on either side of campus. Learn more about [library services and resources](#), and to find research materials in specific subject areas view [the Research Guides](#).

UIC Counseling Center - Counseling services are available for UIC students who pay the student health services fee. You may seek a free and confidential intake with a trained therapist from the Counseling Center, who will work with you to create a clinically appropriate treatment plan. Treatment options may include Brief Individual Therapy or Group Therapy at the Counseling Center, referral to Psychiatry services, or Care Coordination to be connected to a community provider who meets your clinical needs. The Counseling Center is located in the Student Services Building. You may contact them at (312) 996-3490 during business hours to learn more about services, set up an intake, or for crisis intervention. Crisis intervention services are available 24/7 by walking into the Counseling Center during business hours or by calling. If you are calling after business hours, please select “2” when the automated voice message begins. For more information about [the Counseling Center](#) or accessing the Counseling Center and other mental and emotional health services on campus, check out the [Student’s Guide to Accessing Behavioral Health Services at UIC](#). For more information about services in the community, check out our [Community Provider Database](#).

The **Campus Advocacy Network** provides information and offers resources to all UIC students, faculty, and staff. Under the Title IX law, you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim services and advocacy, contact [UIC’s Campus Advocacy Network](#) at 312-413-8206. To make a report to UIC’s Title IX office, email TitleIX@uic.edu or call (312) 996-8670.

Campus Security - As a UIC student, you’ve chosen to live in one of the nation’s largest cities. But, as at any university, crime is a reality. At UIC, we are strongly committed to our public safety programs, and we encourage students to be proactive in learning what programs and services are available in case of an emergency. You are **discouraged** from staying in university buildings alone, including lab rooms, after hours and are **encouraged** to use the [Walking Safety Escort](#) and/or [Night Ride](#) if you are uncomfortable traveling anywhere on campus. Also, you can download [the UIC SAFE app](#), a free personal security tool for students, faculty, and staff. It allows you easy contact with dispatchers and first responders in case of an emergency. Finally, by dialing 5-5555 from a campus phone, you can summon Police or Fire for any on-campus emergency. You may also program the complete number, (312) 355-5555, on speed dial on your phone.

Emergency Response Systems and Guides - The emergency response guides can be found at the [Office of Preparedness and Response](#). Please review and acquaint yourself with the guide and recommendations for various emergency situations.

Student Health Services – If you get sick during the semester, please be aware that there are resources on campus that can support you. The Family Medicine Center is committed to providing high-quality,

*student centered care and services. It has attending physicians, residents, and nurse practitioners on staff to provide the full range of primary health care services. The Family Medicine Center also offers a limited set of health services at **no additional cost to students who pay the Student Health Service Fee (SHSF)**. Services include caring for an acute or urgent illness (e.g., cold, strep throat, asthma attack, flu) or an injury. More information can be found on the [Family Medicine Center's Student Health](#) webpage.*

Disclaimer

This syllabus is intended to give guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance as in-class announcements and in writing via email.

Last Revised: Monday 24 August 2026